Computing Infrastructures

Riassunto completo del corso

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1 Prefazione

1.1 Disclaimer

I contenuti di questo PDF sono stati compilati con il solo scopo di avere una risorsa esaustiva e facilmente consultabile degli argomenti trattati nel corso di Computing Infrastructures sostenuto dal professore Palermo: non sono in alcun modo materiale sostitutivo alle lezioni, che frequentate in presenza mi hanno permesso di capire a fondo la materia. Sconsiglio di usare esclusivamente suddetto documento per cercare di prepararsi all'esame, supporto invece l'uso di questa dispensa per fare il punto della situazione durante il periodo di studio.

1.2 Keywords

- **DC** - Data Center
- **IoT** - Internet of Things
- **AI** - Artificial Intelligence
- **IaaS** - Infrastructure as a Service
- **PaaS** - Platform as a Service
- **SaaS** - Software as a Service
- **LAN** - Local Area Network

2 Datacenters

2.1 Introduction

Datacenter

A Datacenter is a building whose primary purpose is to house a computer room and its support areas, so that services can run continuously and efficiently

There exist lots of different types of datacenters:

- **Enterprise Data Centers** - facilities operated and controlled by the company that provides the service. Main problem: expensive to maintain (to build or not to build dilemma) and can shorten the possibility to generate money by the business.
- **Colocation Data Centers** - larger facilities built to accomodate different businesses and can personalize individual IT parts for individual businesses. Main problem: IT purchases are on your own and restrictions on access and maintenance.
- **Managed Hosting Platforms** - all facilities are managed by third parties, still a single-tenant solution. Main problem: higher operational costs and possibility of customer lock in problem.

In addition to the above solutions, there are also **Cloud Data Centers**, which are the most common solution in post-pandemic times. Those solutions are **IaaS**, **PaaS** and **SaaS**. A clear representation is shown in the figure below. One thing that stands out is that cloud solutions also contain virtualization managed by the provider (you don't know which operating system is running, you are inside an additional abstraction layer).

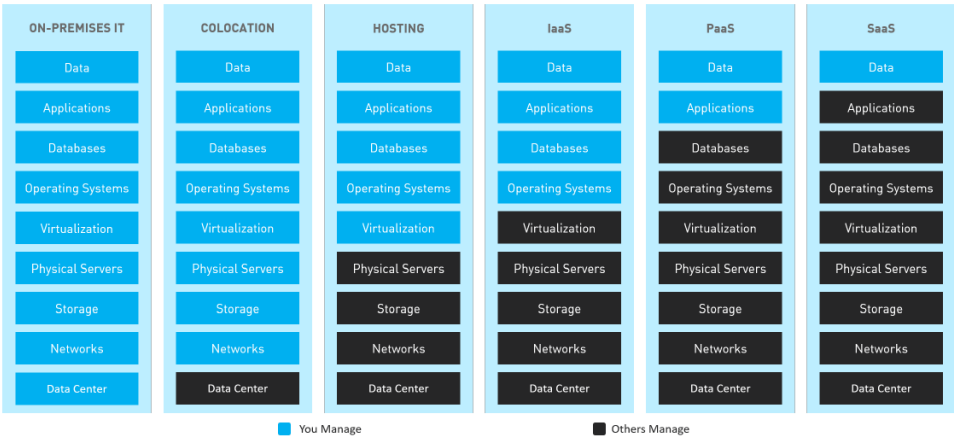

Figure 1: Cloud Data Centers

2.2 Hardware Infrastructures

Let's start with the basic definition of a computing infrastructure.

Computing Infrastructure

A **Computing Infrastructure** is a tecnological infrastructure that provides hardware and software for computation to other systems and services.

Computing infrastructures can be categorized in a pyramid of three layers:

- **IoT and AI-enabled Edge Sensors** - where data acquisition and processing happens at the edge of the network.
- **Edge Servers** - on-premises hardware resources that perform more intensive processing tasks.
- **Data Centers and Cloud** - virtualized computing and capacity on demand.

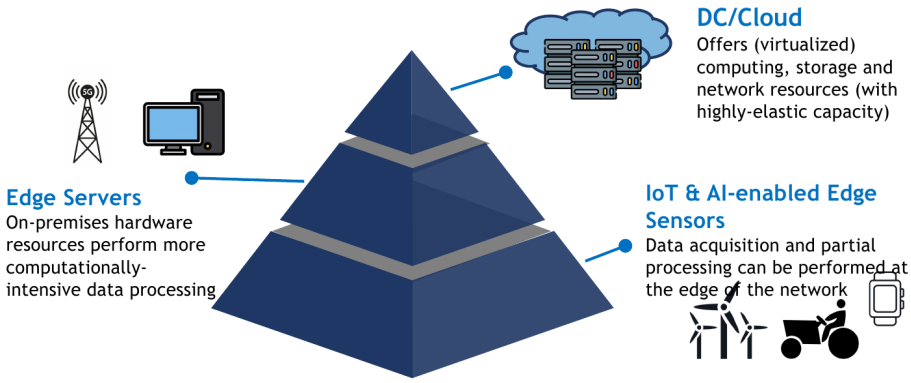

Figure 2: Computing Infrastructure Pyramid

It's obvious that the bottom layer has less memory and computational power, but it is also more responsive and with low latency; the upper layers offer more resources to work with. To categorize all the components of a system it is used what is called the **Computing Continuum**.

Figure 3: Data Centers

2.2.1 Data Centers

Usually, data centers contain servers for **processing**, **storage** and **communication**. They have lower IT costs, high performance, "unlimited" storage capacity, universal data access and have device independence. On the other hand they need a constant internet connection, they do not work well with low speed connections, have high power consumption and cooling costs. Most importantly they add **latency in taking decisions**.

2.2.2 Edge, Embedded and IoT Devices

- **IoT Devices** - useful for their pervasiveness, wireless connection and low costs, sometimes powered by batteries, but with limited resources, security issues and difficult to program.
- **Embedded PCs** - support pervasive computing, with more resources and abstraction layers (programmed as PCs), but with higher costs and power consumption.
- **Edge/Fog Computing Systems** - have high computational capacities, better in terms of privacy, have reduced latency in making decisions, but with higher costs of power and with cloud connection requirements.

What is Edge and what is Fog

In Edge Computing, data is processed at the edge of the network, whereas in Fog Computing data is moved into a LAN before going into a DC.

3 Data Center as a Computer

4 Storage